Unit 3 Guided Notes — Equations & Inequalities

ANSWER KEY

Keep it balanced. Undo step by step. Flip the sign when you divide by a negative.

🎯 **By the end of this unit, I can…**

- Solve multi-step equations, including ones with variables on both sides
- Distribute and combine like terms accurately
- Identify equations with one solution, no solution, or infinitely many

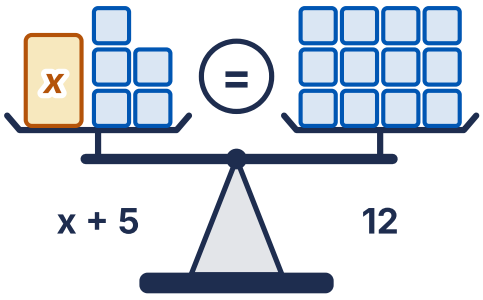
Standards: NC.8.EE.7

Answer key: every blank and final answer is filled in green.

1 One-Step Equations / Inverse Operations

📌 **THE BIG IDEA:** INVERSE OPERATIONS are used to isolate the variable. Whatever you do to one side, you MUST do to the other to keep the equation balanced!

x tile +1 tile red = negative (−x or −1) a big block = that number



x + 5 = 12 is balanced. Keep it balanced!

Operation	Inverse	Example
Addition	Subtraction	x + 5 = 12
Subtraction	Addition	x − 3 = 8
Multiplication	Division	3x = 15
Division	Multiplication	x/4 = 2

Match the Operation to its Inverse

MATCH Write the letter of each operation's inverse on the line.

B	1. Addition	**A.**	Division
D	2. Subtraction	**B.**	Subtraction
A	3. Multiplication	**C.**	Multiplication
C	4. Division	**D.**	Addition

2 Two-Step Equations

📌 **TWO STEPS:** Step 1 — Undo addition/subtraction. Step 2 — Undo multiplication/division.

Example 1: $-2x + 4 = 8$

SOLVE $-2x + 4 = 8$

Step 1: Get rid of the + 4. The opposite of + 4 is − 4, so subtract 4 from BOTH sides.

Step 2: Get x by itself. x is MULTIPLIED by −2. The opposite is dividing, so divide BOTH sides by −2.

Answer: $x = -2$

Example 2: $x/2 + 6 = 7$

SOLVE $\frac{x}{2} + 6 = 7$

Step 1: Get rid of the + 6. The opposite of + 6 is − 6, so subtract 6 from BOTH sides.

Step 2: Get x by itself. x is DIVIDED by 2. The opposite is multiplying, so multiply BOTH sides by 2.

Answer: $x = 2$

3 Distributive Property

📌 **THE RULE:** Any number (or variable) outside the parentheses is multiplied by EVERYTHING inside. Pay close attention to positive/negative signs!

Example: $2(2x + 3) = -6$

SOLVE $2(2x + 3) = -6$

Step 1: Distribute. Multiply 2 by EACH term inside the ( ).

Step 2: Get rid of the + 6. The opposite of + 6 is − 6, so subtract 6 from BOTH sides.

Step 3: Get x by itself. x is MULTIPLIED by 4. The opposite is dividing, so divide BOTH sides by 4.

Answer: $x = -3$

ERROR ANALYSIS **A student expands $-3(x - 4)$ as $-3x - 12$. What went wrong?**

A negative times a negative is positive: $-3 \cdot (-4) = +12$. Correct: $-3x + 12$. Distribute the sign to EVERY term inside the parentheses.

4 Variables on Both Sides

📌 **STRATEGY:** Move variables from the RIGHT to the LEFT using inverse operations, then solve like a two-step equation.

Example: 5x + 3 = 2x + 12

SOLVE **5x + 3 = 2x + 12**

Step 1: Move the x's to the left side. There is + 2x on the right side. Do the opposite: subtract 2x from BOTH sides. Now the x's are only on the left.

Step 2: Get rid of the + 3. The opposite of + 3 is − 3, so subtract 3 from BOTH sides.

Step 3: Get x by itself. x is MULTIPLIED by 3. The opposite is dividing, so divide BOTH sides by 3.

Answer: x = 3

5 Multi-Step Equations — The 5-Step Process

📌 **THE 5 STEPS:** 1. Distribute → 2. Combine Like Terms → 3. Move Variables → 4. Inverse for + or − → 5. Inverse for × or ÷

Example: $-2x + 1 = 8(x + 2) - 5x$

SOLVE $-2x + 1 = 8(x + 2) - 5x$

Step 1: Distribute. Multiply 8 by EACH term inside the ( ).

Step 2: Combine like terms. Put like terms together (x terms with x terms, numbers with numbers): $8x - 5x = 3x$.

Step 3: Move the x's to the left side. There is + 3x on the right side. Do the opposite: subtract 3x from BOTH sides. Now the x's are only on the left.

Step 4: Get rid of the + 1. The opposite of + 1 is − 1, so subtract 1 from BOTH sides.

Step 5: Get x by itself. x is MULTIPLIED by −5. The opposite is dividing, so divide BOTH sides by −5.

Answer: $x = -3$

6 Special Cases — No Solution & Infinite Solutions

🔑 **NO SOLUTION:** Variables cancel and you get a FALSE statement (like 5 = 9).

🔑 **INFINITE SOLUTIONS:** Variables cancel and you get a TRUE statement (like 6 = 6).

Result	Meaning	Example
Variables stay; x = number	One Solution	2x+1=7 → x=3
Variables cancel; FALSE remains	No Solution	2x+1=2x+5 → 1=5 ×
Variables cancel; TRUE remains	Infinite Solutions	2x+1=2x+1 → 1=1 ✓

No Solution — Solve: $3(x + 2) = 3x + 10$

SOLVE $3(x + 2) = 3x + 10$

Step 1: Distribute. Multiply 3 by EACH term inside the ( ).

Step 2: Move the x's to the left side. There is + 3x on the right side. Do the opposite: subtract 3x from BOTH sides. Now the x's are only on the left.

Answer: No solution

Infinite Solutions — Solve: $2(x + 4) = 2x + 8$

SOLVE $2(x + 4) = 2x + 8$

Step 1: Distribute. Multiply 2 by EACH term inside the ( ).

Step 2: Move the x's to the left side. There is + 2x on the right side. Do the opposite: subtract 2x from BOTH sides. Now the x's are only on the left.

Answer: Infinitely many solutions

Quick Check

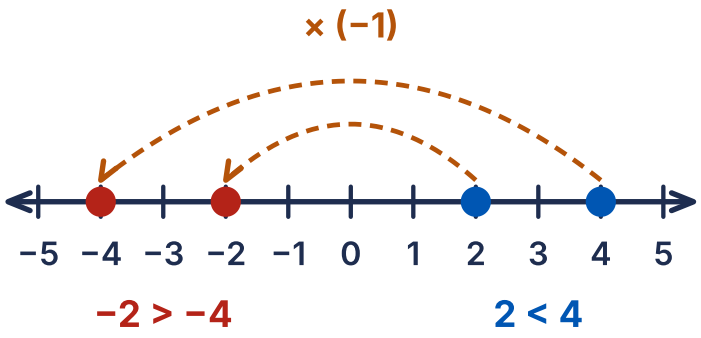
MC Solving an equation gives 7 = 7. The equation has...

- ○ One solution
- ○ No solution
- ● **Infinitely many solutions**

7 Linear Inequalities

Symbol	Meaning	Circle	Shade
<	less than	OPEN	LEFT
>	greater than	OPEN	RIGHT
≤	less than or equal	CLOSED	LEFT
≥	greater than or equal	CLOSED	RIGHT

⚠ **CRITICAL RULE — FLIP THE SIGN:** When you MULTIPLY or DIVIDE both sides by a NEGATIVE number, you must FLIP the inequality sign! Example: −2x > 6 → divide by −2 → x < −3 (sign flipped!)



The order flips, so the sign flips.

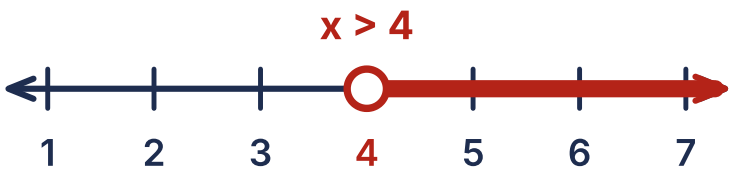
Example (no flip): $2x + 3 > 11$

SOLVE $2x + 3 > 11$

Step 1: Get rid of the + 3. The opposite of + 3 is − 3, so subtract 3 from BOTH sides.

Step 2: Get x by itself. x is MULTIPLIED by 2. The opposite is dividing, so divide BOTH sides by 2.

Answer: $x > 4$



x > 4: open circle (4 is not included)

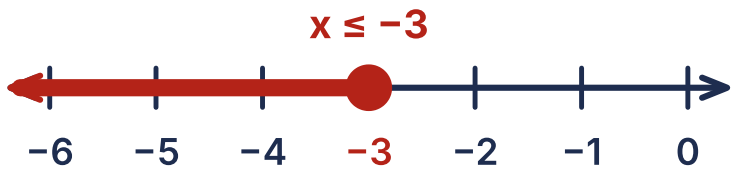
Example (FLIP NEEDED): $-3x + 4 \geq 13$

SOLVE $-3x + 4 \geq 13$

Step 1: Get rid of the + 4. The opposite of + 4 is − 4, so subtract 4 from BOTH sides.

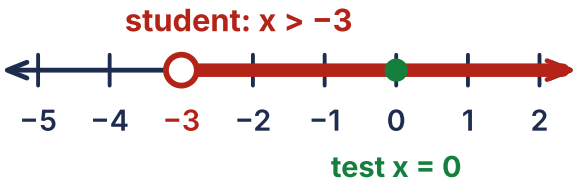
Step 2: Get x by itself. x is MULTIPLIED by −3. The opposite is dividing, so divide BOTH sides by −3. −3 is negative, so the inequality sign FLIPS.

Answer: $x \leq -3$



x ≤ −3: closed circle (sign flipped!)

ERROR ANALYSIS To solve −4x > 12 a student wrote `x > −3`. What's the mistake?



Test x = 0 in −4x > 12. Is −4(0) > 12 true?

Dividing both sides by the negative number −4 requires FLIPPING the sign. Correct: x < −3.

8 Word Problems — Equations & Inequalities

📌 **THE FLOWCHART:** 1) READ → 2) DEFINE the variable → 3) IDENTIFY keywords → 4) WRITE the equation/inequality → 5) SOLVE → 6) CHECK → 7) ANSWER in a sentence.

Equation (=)	Inequality	Addition	Subtraction
total, equals, is, the same as, gives, results in	at least (≥), at most (≤), more than (>), fewer than (<)	sum, plus, more than, increased by, total	difference, minus, less than, decreased by

SHORT ANSWER Movie Theater: A movie theater charges a $6 entry fee plus $4 per ticket. If the total cost was $30, how many tickets were purchased?

Let x = number of tickets

Equation: 6 + 4x = 30

Solve (show each step):

Answer: 6 tickets were purchased.

SHORT ANSWER Notebooks: A student wants to spend no more than $50 on notebooks that cost $6 each. How many notebooks can the student buy?

Let x = number of notebooks

Equation: 6x ≤ 50

Solve (show each step):

Answer: x must be a whole number, so the student can buy **at most 8 notebooks.**

✏ Part 2 — Practice Problems

📌 **Show every step on its own line.** Box your final answer.

A. Multi-Step Equations

1) $3x + 7 = 22$

SOLVE $3x + 7 = 22$

Answer: $x = 5$

2) $2(x - 4) = 10$

SOLVE $2(x - 4) = 10$

Answer: $x = 9$

3) $5x - 3 = 2x + 9$

SOLVE $5x - 3 = 2x + 9$

Answer: $x = 4$

4) $-4(x + 2) = 12$

SOLVE $-4(x + 2) = 12$

Answer: $x = -5$

5) $6x + 4 = 4x - 8$

SOLVE $6x + 4 = 4x - 8$

Answer: $x = -6$

6) $3(2x + 1) = 4x + 7$

SOLVE $3(2x + 1) = 4x + 7$

Answer: $x = 2$

7) $x/4 + 5 = 8$

SOLVE $\frac{x}{4} + 5 = 8$

Answer: $x = 12$

8) $7 - 2x = 3x + 17$

SOLVE $7 - 2x = 3x + 17$

Answer: $x = -2$

9) 0.5x + 3 = 7

SOLVE $0.5x + 3 = 7$

Answer: x = 8

10) −(x − 6) = 2x + 3

SOLVE $-(x - 6) = 2x + 3$

Answer: x = 1

B. Special Cases — No Solution or Infinite Solutions?

1) 2x + 3 = 2x + 3

SOLVE $2x + 3 = 2x + 3$

Answer: Infinitely many solutions

2) 4x + 1 = 4x − 5

SOLVE $4x + 1 = 4x - 5$

Answer: No solution

3) 3(x + 2) = 3x + 6

SOLVE $3(x + 2) = 3x + 6$

Answer: Infinitely many solutions

4) 5x − 5 = 5(x + 1)

SOLVE $5x - 5 = 5(x + 1)$

Answer: No solution

5) 7x + 2 = 7x + 2

SOLVE $7x + 2 = 7x + 2$

Answer: Infinitely many solutions

6) x + 4 = x − 1

SOLVE $x + 4 = x - 1$

Answer: No solution

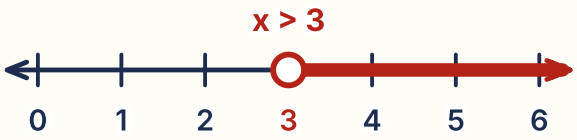
C. Inequalities — Solve

Solve it like an equation. FLIP the sign only when you multiply or divide both sides by a negative number.

1) 3x + 2 > 11

SOLVE **3x + 2 > 11**

Answer: x > 3

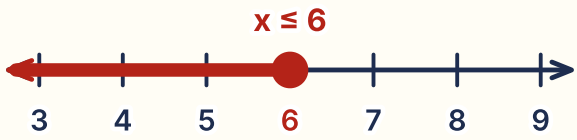


x > 3: open circle at 3, shade right

2) 2x − 5 ≤ 7

SOLVE **2x − 5 ≤ 7**

Answer: x ≤ 6

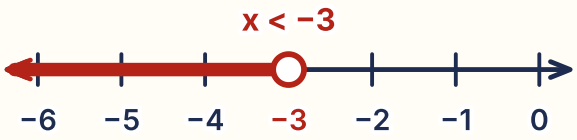


x ≤ 6: closed circle at 6, shade left

3) −4x > 12 (flip!)

SOLVE **−4x > 12**

Answer: x < −3

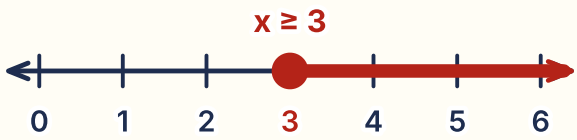


x < −3: open circle at −3, shade left

4) 5x + 1 ≥ 16

SOLVE **5x + 1 ≥ 16**

Answer: x ≥ 3

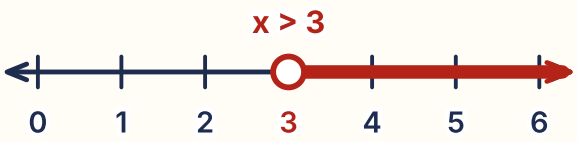


x ≥ 3: closed circle at 3, shade right

5) $7 - x < 4$ (flip!)

SOLVE $7 - x < 4$

Answer: $x > 3$

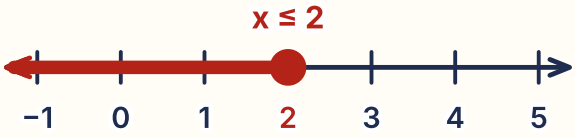


x > 3: open circle at 3, shade right

6) $2(x + 3) \leq 10$

SOLVE $2(x + 3) \leq 10$

Answer: $x \leq 2$

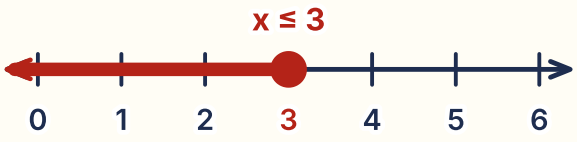


x ≤ 2: closed circle at 2, shade left

7) $-3x + 9 \geq 0$ (flip!)

SOLVE $-3x + 9 \geq 0$

Answer: $x \leq 3$

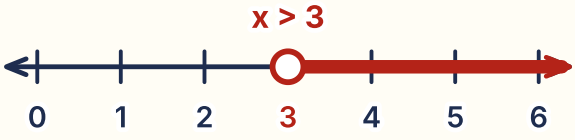


x ≤ 3: closed circle at 3, shade left

8) $4x - 1 > 2x + 5$

SOLVE $4x - 1 > 2x + 5$

Answer: $x > 3$



x > 3: open circle at 3, shade right

D. Word Problems

1. A phone plan costs $35/month plus $0.10 per text. Last month's bill was $52. How many texts were sent?

Let x = number of texts

Equation: 35 + 0.10x = 52

Solve (show each step):

Answer: 170 texts were sent.

2. Sam has $240 and saves $30/week. Jake has $420 and spends $10/week. When will they have the same amount?

Let x = number of weeks

Equation: 240 + 30x = 420 − 10x

Solve (show each step):

Answer: They will have the same amount after 4.5 weeks.

3. A rectangle's perimeter is 56 cm. The length is 4 cm more than the width. Find the width.

Let x = width (cm), so the length is x + 4

Equation: 2(x + 4) + 2x = 56

Solve (show each step):

Answer: The width is 12 cm.

4. Movie tickets cost $9 each. You have at most $50 to spend. What is the maximum number of tickets?

Let x = number of tickets

Equation: 9x ≤ 50

Solve (show each step):

Answer: x must be a whole number, so you can buy **at most 5 tickets.**

5. A car rental costs $45/day plus $0.25/mile. Your budget is $120 for one day. How many miles can you drive?

Let x = number of miles

Equation: 45 + 0.25x ≤ 120

Solve (show each step):

Answer: You can drive **at most 300 miles.**

6. Three consecutive even integers sum to 78. Find the smallest integer.

Let x = the smallest even integer, so the next two are x + 2 and x + 4

Equation: x + (x + 2) + (x + 4) = 78

Solve (show each step):

Answer: The smallest integer is **24** (the three numbers are 24, 26 and 28).

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